

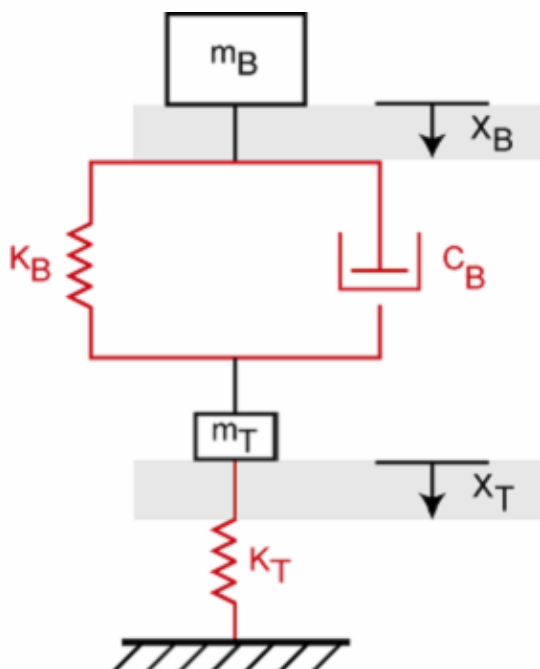
Race car Damping – Some food for thought

The first article I ever wrote for Racecar Engineering was how to specify dampers using a dual rate damper model. This was an approach that my colleagues and I had applied with considerable success. However at the time I wrote the article I said this was definitely not the last word on race car damping. This article will carry on this discussion.

This will be the first in a series of articles on race car damping. I'm doing this for a number of reasons. Firstly to try and cover the whole panacea of race car damping in one article is lunacy. It is simply too big. The second reason is that at the time writing I don't have all the answers either. However what I do want to do with this article is set some food for thought and really encourage a good public discussion and hopefully bring a few people out of the wood work.

On a more philosophical note one of the greatest tragedies I see in our line of work is that it is very rare to have a genuine technical discussion. This is as a direct result is that we are either so focused on the business of winning we actually forget what we can learn from others. It's also a direct result that some times we come up with a problem at the race track that we can't solve straight away so we have to fudge a solution. Consider this the start of one of these genuine technical discussions that can really delve into what is going on. Also be warned this will not be one of these discussions that are going to be this is the problem and this is the answer.

To kick this off Let's consider our quarter car model. I've stated on a number of different occasions that this is the basic building block of car suspension analysis. It's also really simple to quantify. The quarter car model is presented in Fig-1,



The parameters are

m_B = Mass of the sprung mass (kg)

m_T = Mass of the unsprung mass (kg)

K_B = Sprung mass spring rate (N/m)

C_B = Sprung mass damping (N/m/s)

K_T = Tyre spring rate (N/m)

Fig-1 – Quarter car model

Before I dive into the equations of motion let me just be clear about the nomenclature,

x_b	= Movement of the sprung mass (m)
x_w	= Movement of the wheel (m)
A_R	= Road amplitude (m)
f	= Frequency of road input (Hz)

We are going to subject our quarter car model to a sinusoidal input of amplitude A_R and frequency f . I am choosing this for two reasons. The first is for simplicity. I want to layout the obvious case before going into further detail. Also thanks to our good friend Joseph Fourier repeating signals can be summed as a bunch of sine waves. Consequently this forms a very good starting point.

The equations of motion for our case study here are the following,

$$\begin{aligned} m_B \cdot \ddot{x}_B &= -K_B \cdot (x_B - x_w) - C_B \cdot (\dot{x}_B - \dot{x}_w) \\ m_w \cdot \ddot{x}_w &= K_B \cdot (x_B - x_w) + C_B \cdot (\dot{x}_B - \dot{x}_w) - K_T \cdot (x_w - A_R \cdot \sin(2\pi f \cdot t)) \end{aligned} \quad (1)$$

Just to make this discussion simpler I'm considering the system from it's equilibrium state. That is the zero state is with the quarter car model on the ground and at rest. Or to use data engineer's parlance we have zeroed it on the ground. It's the reason I haven't included the mass terms in equation (1). Bare with me it will become clear why I have done this in a few paragraphs.

Let's consider the case what should the damping be if we want the body to remain at the rest position. That is what should we do to try and duplicate active suspension. This has some rather interesting ramifications. These are summarised in the following equations,

$$\ddot{x}_B = 0 \quad (2)$$

$$x_B - x_w = A_R \cdot \sin(2\pi f \cdot t)$$

$$\dot{x}_B - \dot{x}_w = 2\pi f \cdot A_R \cdot \cos(2\pi f \cdot t) \quad (3)$$

$$\ddot{x}_B - \ddot{x}_w = -(4\pi^2 f^2) \cdot A_R \cdot \sin(2\pi f \cdot t)$$

Equation (3) represents the situation that the measured damper movement is effectively the suspension system riding the bump profile (here a sine wave) perfectly.

So using equation (1) to (3) we can now solve for what we would like the damping to be. Plugging in the numbers we can show the damping rate as a function of time should look something like this,

$$c_B = -\frac{K_B}{2\pi f} \cdot \tan(2\pi f \cdot t) \quad (4)$$

Hooray! We can have the perfectly damped car. There's just one small problem, which the mathematicians amongst will instantly recognise (See trigonometry and trig functions weren't just devised as creative intellectual water torture). This small problem I'm going to illustrate graphically. This is illustrated in Figure 2.

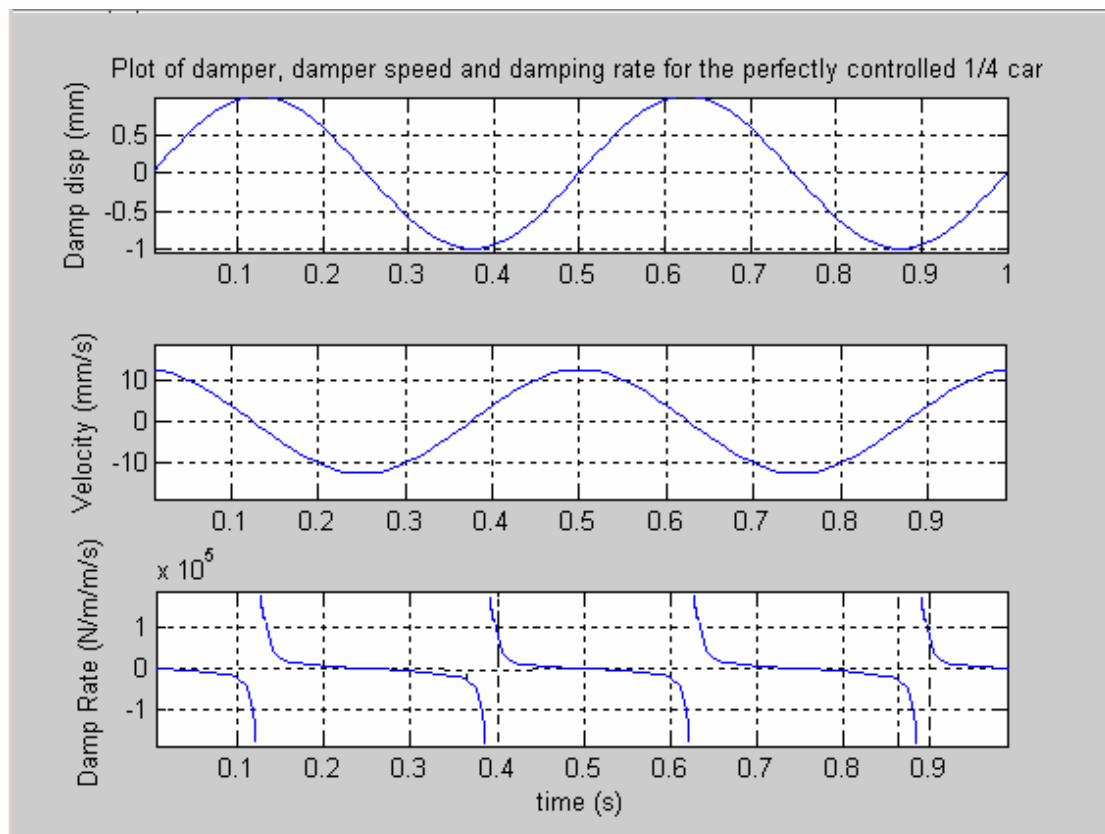


Fig-2 – Illustration of damper position and the damping rate required for perfect body control.

What's happening here that as the bump amplitude hits it's maximum and the damper velocity hits it's 0 the damping rate goes infinite. This represents the fact that we are trying to get the damper to do what effectively isn't a damper job. This arises because we are trying to get the damper to hold position when it has no influence to do so.

However all is not completely lost. We do have some options. Option 1 is to pray to what ever deity your worship for an infinite damping rate. Some how this might be a prayer that goes unanswered. Option 2 is to approximate this with a max damping rate and blow it off very quickly. The best way to approximate this is to plot damping rate vs damping velocity and integrate the damping rate to get the damper force. (Hint motorsport engineering students and young data engineers this is a job for you). You should get something that looks like Fig-3,

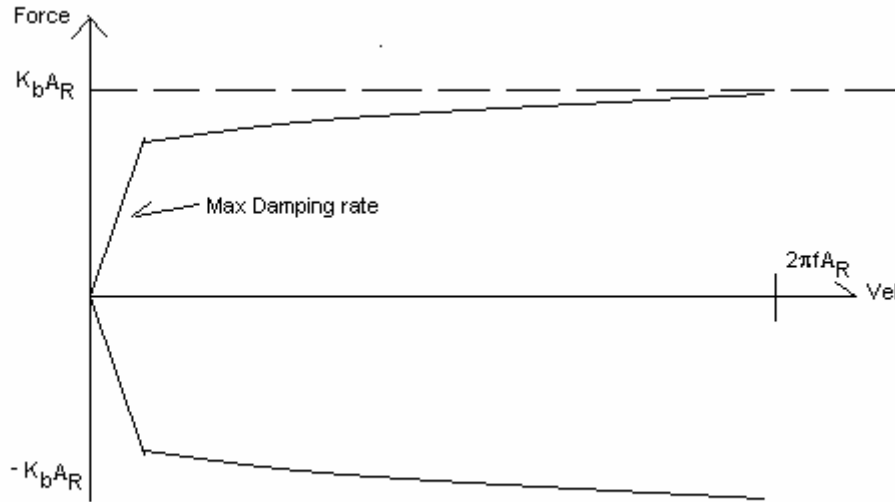


Fig-3 – Idealised damping curve.

This could have half a chance there is just one small problem, the high initial damping rate is effectively spring pre load. As we have discussed in previous articles this can potentially have disastrous consequences as the car loads up in braking and cornering.

The only pay off of this is it gives a way off looking at a frequency analysis of the data from track to track and piecemealing damping rates for various damper positions for each particular circuit. It's far from perfect but it's an idea worth investigating.

The other approach we might like to canvas is dispensing with the damper entirely and replace it with an inerter. Let's define the inertance as a value in b in kg and place the inerter between the wheel and the body. In this case the governing equation at the body reduces to,

$$m_B \cdot \ddot{x}_B = -K_B \cdot (x_B - x_w) - b \cdot (\ddot{x}_B - \ddot{x}_w) \quad (5)$$

Using equation (3) and asking for the body not to move it can be seen that,

$$b = \frac{K_B}{(2\pi f)^2} \quad (6)$$

These terms cancel out because the sine term is both in position and acceleration. What is physically happening is that the inerter is anticipating what is coming and can make the necessary adjustments in the force.

At face value it would appear we have reached the promised land. However as always the devil is in the detail. First things first at low frequencies the value of the inerter is going to be rather large, almost to the point where it is impractical to use. The second thing is that this will work really well for the specified frequency. There is no guarantee at what happens outside this specified frequency.

What all this illustrates is that passive elements don't truly replace a full active suspension system. I personally think the banning of active suspension in most level of motorsports is one of the greatest tragedies to be visited on our sport. What I would love to be writing about is how we can use active suspension to truly optimise what a car is capable of. However the powers at be have seen fit to stick us with what we have.

This also illustrates why race engineers are always playing with damper adjustments. As can be seen very clearly what works for one condition will not necessarily work for another. This is why race teams spend money on simulation and other tools so they can get some idea of what to do when they go from track to track. Also as illustrated by this example sometimes it can be a challenge to see the forest through the trees.

To put this discussion in it's proper context, let's actually review the dual rate approach I discussed in my first article. This approach actually had its origin in control system analysis, and analysing the fundamental equations of motion to illicit a desired response. That is heavily damped to control the body motion and lightly damped to allow the bumps to blow off.

The heart of this approach was to evaluate the natural frequency of the quarter car model and then use this to evaluate the damping ratio. For the readers convenience I'll present the two equations below, which are the heart of the technique,

$$\omega_0 = \sqrt{\frac{K_B}{m_B}} \quad (7)$$

$$C_B = 2 \cdot \omega_0 \cdot m_B \cdot \zeta \quad (8)$$

This is the first point of the analysis. Equation (7) specifies the natural frequency in rad/s of the system, and equation (8) means a damper rate based on the damping ratio that is desired can be specified. Please note the spring rate is specified in N/m and the mass is specified in kg. What this means is that we can now specify a damping rate based on what we want the system to do as opposed to guessing. Also because of the form of equation (8) this can be readily applied for both the front and the rear. How to choose the damping ratio is illustrated in Fig-4,

Region	Value
$\zeta_{LS-Bump}$	This is body control so, if it is a bumpy circuit the ratio is 0.5. If the circuit is smooth choose 0.7. If the tyres need to be worked hard then choose 1.2.
$\zeta_{LS-Rebound}$	This will be dependant on a number of factors. Where body control is not significant use 0.3. However where body control is significant use 0.5 to 0.7.
$\zeta_{HS-Bump}$ $\zeta_{HS-Rebound}$	This is the high pass area of the damping curve. The values here should be 0.3 to 0.4

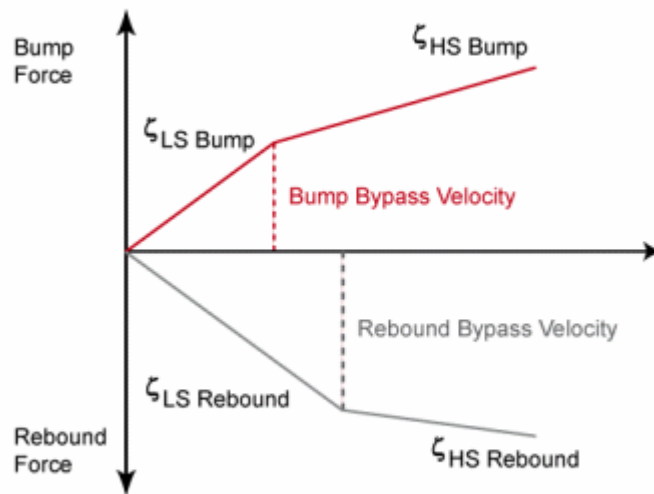


Fig-4: Damping ratio selection guide

For the derivation of this I will refer the reader to my article on Specifying Dampers.

Now that we have reminded ourselves of this method how does this stack up in controlling our quarter car model rolling down the road. To illustrate this graphically consider Fig-5,

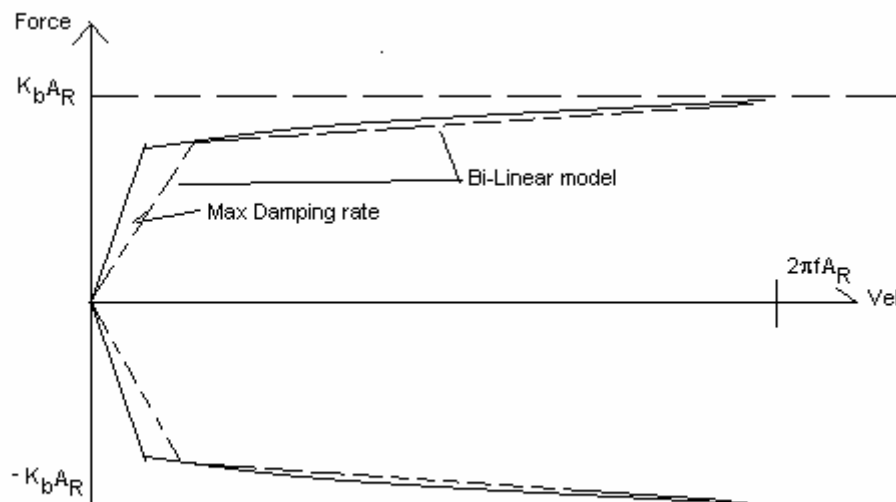


Fig-5 – Comparison of Calculated damping vs using a bilinear model.

For ease of comparison I've illustrated the bilinear rate model using a dashed line and the calculated line as solid. It can be clearly seen that using the bi linear approach while we don't get the exact response where after we can get pretty close.

Now before some of you start howling in protest there is a precedent for my little party trick. For subsonic aircraft the optimum wing planform is an ellipse. The most marked example of this is the Supermarine Spitfire of World War II fame. However a really easy approximation of this is to do a straight chord for half the wing span and then at the tips the chord is half the length of the centre of the wing. To better illustrate this look at Figure 6,

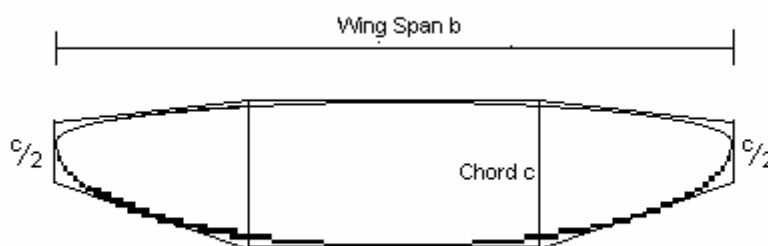


Fig-6: Straight-line approximation of an ellipse

This gets you pretty close performance wise with something that is really easy to manufacture. This configuration is seen on thousands of light aircraft the world over. The moral of the tale is don't underrate a few strategically placed straight lines.

The other advantage of the damping ratio approach is it actually makes it really easy to quantify what is going on with the car. In my article on advanced aerodynamic analysis I walked the reader through on State space analysis and the use of eigenvalues and eigenvectors. This involved this horrible mathematical monster called complex numbers (Well it's actually not all that horrible it is actually a mathematician's fudge to the conundrum of $i^2 = -1$, but the fudge actually worked quite well) but the use of these complex numbers actually well tell you what the car is going to do.

In closing let's just reflect on a number of things. First off it's impossible to get a passive suspension system to do what an active system can. You can get away with it in certain cases, but in all honesty you wouldn't bet your house on it. Secondly our old war horse of using a linear approximation based on damping ratios can be stretched. Yes it's not perfect, but sometimes getting something in the ball park is sometimes good enough. But more importantly, for the time being I'm turning the discussion over to you.

As I stated in the beginning of this article this will be the first of many articles on this matter. This article has set the tone by outlying the challenge we face by considering a rather simple example

that illustrates really well what we are up against. Until the next instalment I turn this over to you the reader to reflect on this and see what you can come up with. If this is the only result that comes of this then I have succeeded.